FBCSP-based Multi-class Motor Imagery Classification using BP and TDP features

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***Abstract*— Use of Motor Imagery in EEG signals is gaining importance to develop Brain Computer Interface (BCI) applications in various fields ranging from bio-medical to entertainment. Filter Bank Common Spatial Pattern (FBCSP) algorithm is a promising feature extraction technique to deal with subject-specific behavior in Motor Imagery classification. Using FBCSP on EEG we have developed an accurate but less computationally expensive approach by making use of Time Domain Parameters (TDP) and Band Power (BP) features to form a combined feature set. The novelty of our approach is also the use of optimal time segmentation to overcome non-stationary state behavior of Event-Related Desynchronization (ERD) and Event-Related Synchronization (ERS) over time. We analyzed the impact of parameter variations on classification accuracy and achieved 0.59 mean kappa value for Dataset 2a BCI competition IV, the highest reported for FBCSP approaches, along with the lowest inter-subject variation.**

I. INTRODUCTION

Brain Computer Interface (BCI) based on Motor Imagery (MI) classification in Electroencephalograph (EEG) can be used to assist the amputees and people who can not control their limbs because of neurological disorder. For such purposes, Common Spatial Pattern (CSP) is used as optimal spatial filters to extract features for classification of EEG signals for BCI applications. It discriminates two classes data into additive sub-components by maximizing variance between the classes [3]. A variant of CSP was proposed as Local Temporal Common Spatial Pattern (LTCSP) [4]. Time-dependent neighbouring matrix was used to make co-variance matrix less prone to noise. They used OVR method and LDA classifier on the features obtianed form LTCSP, which returned 0.61 mean kappa value for four class EEG data. In [5], Researchers proposed optimized spatial-frequency-temporal feature extraction method. They selected subset of channels using Fisher radio and then decomposed EEG signals into time-frequency components. They applied CSP algorithm on time-frequency EEG data and obtained spatial-frequency-temporal feature after applying sparse regularization. They used Nave Bayesian classifier and obtained 0.53 mean kappa value. For multiclass problem, [6] introduced Filter Bank Common Spatial Pattern (FBCSP) to select (autonomously) key temporal-spatial discriminative characteristics of electroencephalograph data. FBCSP can maximize variance in all classes unlike CSP which is used for binary classes. In this paper, researchers tried different classification methods for multi-class data i.e. Pair-Wise (PW), Divide and Conquer (DC), One-Versus-Rest (OVR) and Naive Bayesian Parzen Window (NBPW) and achieved 0.57 mean kappa value by using FBCSP with OVE and PW classification method.

Band Power (BP) features can be used to mine hidden features in a subject-specific time-frequency analysis as shown in recent evidence [7] [15]. In the literature there is a surprising number of publications which used Time Domain Parameters (TDP) to mine temporal features [8] [9] [10] [15]. In [8], TDP is used in terms of activity, mobility, and complexity to compute velocity and other features in extension and flexion positions of an index finger, middle finger and thumb. On account of these facts, we used BP and TDP as integrated features to explore subject specific and temporal information as combined information hidden in EEG signals. Many experts regard subject specific time segment and frequency band selection as still a challenging task. In this study, we came up with near-optimal parameters and proposed solution for the aforementioned challenge by selecting appropriate time segments, frequency bands, smoothing window for BP, the order of derivative for TDP and feature window. Further, we formed the final feature set by combining BP and TDP parameters extracted using FBCSP. To the best knowledge of authors, BP and TDP are not used before as individual or combined feature set with FBCSP for the dataset 2a, BCI compettition IV.

II. PROPOSED METHOD

The proposed algorithm with Filter Bank Common Spatial Pattern (FBCSP) for feature extraction is shown in Fig. 1. In first preprocessing stage, artifacts reduction, time segmentation and selection of frequency bands are performed. In FBCSP stage, we divided these bands into multiple sub-bands and CSP filters are applied to all individual sub-bands. Most discriminative feature are selected and fed to feature computation stage where BP and TDP are used to form final feature set for classifier.

A. *Preprocessing*

Preprocessing stage is used for artifacts reduction, frequency bands (μ and β) selection and time segmentation. In our case we are interested in ERD/ERS phenomena which occurs in μ and β frequency bands and also in corresponding time segments in which the actual motor imagery phenomena has occurred.

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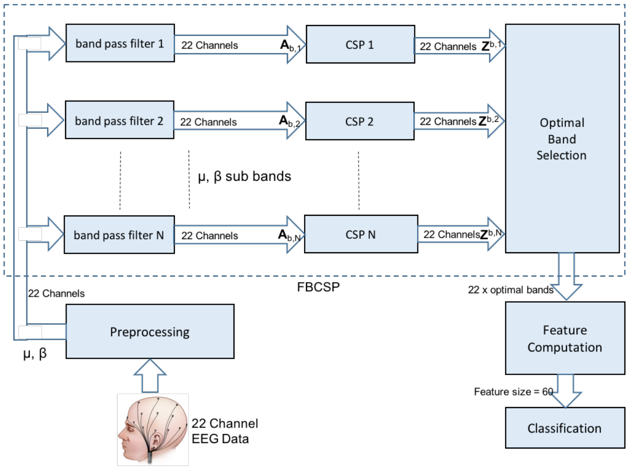

Fig. 1. Proposed Model

1) Frequency Bands Selection: Artifacts present in EEG data can result as decrease in classification accuracy. To remove artifacts and appropriate frequency band selection related to MI phenomena, Butterworth filter of 5^th order has been used. As MI phenomena occurs between 8-30 Hz, μ(8-13 Hz) and β(14-30) bands are used and the remaining frequency ranges are dropped.

2) Time Segmentation: In each trial, subjects were asked to imagine motor execution from the time stamp of 3.0-6.0. Timing window of actual motor imagery phenomena varies person to person and can occur in different duration for all individual subjects. We experimented with the time segment parameter in this study and performed time segmentation of $3.5-6.0$ sec and $4.0-6.0$ sec on average for BP and TDP features respectively.

B. Feature Extraction

For feature extractin, Filter Bank Common Spatial Pattern (FBCSP)[6] is used to maximize dissimilarity between all classes which contributes towards the final feature set. Features are generated after passing all the trials from FBCSP as shown in Fig. 1. FBCSP algorithm comprises of three progressive stages of band-pass filtration: multiple bands selection, spatial filtration using CSP, and feature selection. After preprocessing, each band (μ,β) contains 22 channel data and divided into N non-overlapping subbands using band pass filters of FBCSP block.

CSP algorithm is used to perform spatial filtration on each subbands which increase the separability of binary classes by transforming the EEG signals in another domain which maximizes variance for one class and at the same time minimizes the variance for other classes. This is given as:

$$\mathbf{Z}^{b,i} = \mathbf{W}_b^T \mathbf{A}_{b,i} \tag{1}$$

where $\mathbf{A} \in R^{c\times t}$ is single i^{th} trail of motor imagery signal fed as input to the CSP filter (CSP_i), $\mathbf{Z_{b,i}} \in \mathbf{R^{c\times t}}$ denotes filtered output of CSP_i and $W_b \in R^{c\times c}$ is CSP projection matrix. Where 'c' and 't' represent the channels and number of samples per channel respectively. $\mathbf{W}_b$ produce features for which variance among two classes is maximum. It can be obtained using Eigenvalue problem as:

$$\sum_{b,1} \mathbf{W}_b = (\sum_{b,1} + \sum_{b,2}) \mathbf{W}_b \mathbf{D}_b \tag{2}$$

where $\sum_{\mathbf{b,1}}$ and $\sum_{\mathbf{b,2}}$ are estimated covariances and $\mathbf{D_b}$ is diagonal matrix of eigenvalues. The n pairs of features of the i^{th} trial for the b^{th} band-pass filtered EEG signal obtained using CSP are then given by:

$$\mathbf{V}_{b,i} = log \frac{diag(\bar{\mathbf{W}}_b^T \mathbf{A}_{b,i} \mathbf{A}_{b,i}^T \bar{\mathbf{W}}_b)}{trace[\bar{\mathbf{W}}_b^T \mathbf{A}_{b,i} \mathbf{A}_{b,i}^T \bar{\mathbf{W}}_b]} \tag{3}$$

where $\mathbf{V_{b,i}} \in \mathbb{R}^{2n}$. First n and last n columns of $\mathbf{W_b}$ are represented by $\bar{\mathbf{W}}_\mathbf{b}$. Where numerator and denominator are square matrices. $diag$ returns a diagonal vector of this square matrix and $trace$ returns the sum of this vector over each element. Optimal band selection block selects the bands from projected bands (in spatial domain) that have maximum variance which in our case has been selected as two.

C. Feature Computation

Initial Feature set obtained from FBCSP is used to compute final features (spatial and temporal features) using Band Power (BP) and Time Domain Parameters (TDP). Using only band power is not enough to discriminate the EEG classes where as temporal information fused with spatial features could discriminate EEG classes in a better way which is the motivation of combining BP and TDP features in this research. They are collectively able to deal with the non-stationary nature of ERD and ERP events. Both feature selection techniques are described in the following sections.

1) Band Power (BP): Band power (BP) measure average power and instantaneous power present in a trial and divide this difference by average power for normalization or each individual trial [14]. BP feature can be obtained using following equations:

$$\vec{P}(t) = \vec{V}^2(t) \tag{4}$$

Where $\vec{P}$ is the time course of power and $\vec{V}$ is the input. Smoothing window is used to smooth the BP features. Hence onset and exact motor imagery action vary person to person so near optimal parameter of smoothing window is used. Average power can be calculated as:

$$P_{avg}(\tau) = \frac{1}{M} \sum_{i=1}^{M} \vec{P}(t_i + \tau) \tag{5}$$

where i $\in$ 1,2,...,M.τ represent the time of a trial and average power is represented by $\vec{P}_{avg}(\tau)$. Event-Related Desynchronization (ERD) is short-lasting attenuation in μ and β bands which measures absolute mean of the ratio between power difference (instantaneous and average) and average power as:

$$\mathbf{F}_{BP} = mean\Big(\Big|\frac{\vec{P}(t) - P_{avg}(\tau)}{P_{avg}(\tau)}\Big|\Big) \tag{6}$$

Where $\mathbf{F}_{BP}$ is ERD feature set obtained using instantaneous power ($\vec{P}(t)$) and average power (P_{avg}) which is calculated against per frequency band per time segment.

TABLE I
SUMMARY OF OPTIMAL PARAMETERS

	SVM		LDA	
	BP	**TDP**	**BP**	**TDP**
Frequency Bands (Hz)	8-14 20-26	8-15 19-24	8-14 20-26	8-15 19-24
Time Segments (s)	3.5-6.0	4.0-6.0	3.5-6.0	4.0-6.0
Smoothing Window	2.0	-	2.0	-
Filter Type	BWF	BWF	BWF	BWF
Degree of Derivative	-	5^{th}	-	5^{th}
Kernel Width	100	100	-	-
Soft Margin	100	100	-	-
Classifier Mode	OVR	OVR	OVR	OVR

a) Smoothing window: Averaging Band Power (BP) over an entire trial could increase computational cost. to reduce computational cost, smoothing window of size 2.0 sec is used for averaging BP [14].

2) Time Domain Parameters (TDP): TDP is another feature which is used in contrast to BP to exploit maximum required information hidden in EEG trials. To obtain time domain parameters, the variance of power is calculated with the first near-optimal k number of derivatives as:

$$\vec{P}_i(t) = var\left(\frac{d^i\vec{V}(t)}{dt^i}\right) \tag{7}$$

$$\vec{Q}_i[n] = \omega.\vec{P}_i[n] - (1-\omega)\vec{Q}[n-1] \tag{8}$$

$$\mathbf{F}_i^{TDP}[n] = ln(\vec{Q}_i[n]) \tag{9}$$

$$\mathbf{F}^{TDP} = [\mathbf{F}_1^{TDP}, \mathbf{F}_2^{TDP}, ..., \mathbf{F}_N^{TDP}] \tag{10}$$

Where $i = (0, 1, 2, 3, ..., N)$, $\vec{P}_i[n]$ is sampled version of $\vec{P}_i(t)$ and $\vec{P}_i(t)$ is ith derivative of input signal $V(\vec{t})$, $\vec{Q}[n]$ is flatted output of IIR filter, n is number of samples (4) and ω is the parameter for moving average window.

a) Derivative Order: The derivative order has an impact on TDP features. We selected the order of derivative between 1 and 5 as near-optimal order of derivative corresponding to maximum classification accuracy.

D. Feature Fusion

From previous stage of Feature Computation, we obtained BP and TDP features as F_{BP} and F_{TDP} respectively which are used as the final feature set after concatenation in a single feature vector. We obtained 24 features form BP and 36 feature form TDP and final feature set contain 60 $(24+36)$ features which are used to train the classifiers.

E. Classification

We tried two different classifiers to train the model with our proposed feature extraction technique. Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM) are used separately for training. Ground truth is labeled in numbers $(1, 2, 3, 4)$ for four classes of MI data. Hence both classifiers are binary classifiers, to classify multi-class data, LDA classifier is used in One-Versus-Rest (OVR) mode whereas SVM is used with a Gaussian kernel.

F. Dataset Configuration

We used BCI Competition IV [11] Dataset 2a (approved by Institutions Ethical Review Board), available publicly, which is comprised of four classes (left, right, feet and tongue) of motor imagery measurements collected from 9 subjects. Dataset recorded in 2 sessions, one for training and one for testing and evaluation. there are total 288 trails recorded using 22 EEG channels and 3 EOG channels.

III. EXPERIMENTS

A time segment of 3.5-6.0 is extracted just after the onset of visual cue. For testing, a complete trail is passed to trained model without omitting any data from a single trail.

A. Evaluation Benchmark

Cohen's Kappa (k) is used as evaluation matrix. It is recommended for evaluation by BCI competition IV committee. k is a robust statistics used for testing of reliability of intrarater or interrater and can be formulated as

$$k = \frac{\sum \pi_{ii} - \sum \pi_{i+}\pi_{i+}}{1 - \sum \pi_{i_+}\pi_{i_+}} \tag{11}$$

where π_{ii} is the probability of occurrence of two events in a same category i, π_{i+} is agreement by chance, and $\sum \pi$ is the total probability of the intrarater and interrator agreement, computed from the confusion matrix. $k = 0$ represent the amount of agreements occurred by chance (randomly) and $k = 1$ represent full agreement whereas $k \leq 0$ results if agreement becomes less than the agreement by chance.

B. Results and Discussion

We tried an extensive number of experiments to identify which combination of parameters perform better with LDA and SVM classifiers. Table I summarizes the optimized parameters.

Accuracy comparison of proposed method with other techniques is given in Table II. Winner of BCI Competition IV for Dataset 2a [13] used FBCSP as feature extraction and mutual information method for feature selection. They achieved 0.57 MKV and 0.183 as standard deviation using their modified feature extraction approaches: Mutual Information-based Best Individual Feature (MIBIF) and Mutual Information-based Rough Set Reduction (MIRSR) with parson window. In [12], researchers were able to achieve 0.58 MKV where they obtained standard deviation of 0.268. As compared to FBCSP approaches, proposed features (BP and TDP) combined with FBCSP, outperformed the reported methods in literature with both LDA and SVM classifiers [12] [6] [5] [16] by yielding 0.59 MKV and 0.124 standard deviation.

In [4], researchers used LTCSP and achieved 0.61 MKV but still have a greater standard deviation of 0.196. Though we have not tried LTCSP yet, even with FBCSP the BP and TDP features has resulted in a comparable accuracy (0.58 mkv) and shows less variation in performance over different subject. Also computational complexity of proposed model is lower than the best reported method which is discussed further as follows:

TABLE II

CLASSIFICATION ACCURACY COMPARISON (KAPPA COEFFICIENTS) ON BCI COMPETITION IV DATASET 2A, KEY: MKV: MEAN KAPPA VALUE

	Proposed method	**Raza, H.**	**Ang, Kai Keng**[6]			**Miao, Minmin**	**Ghabri, H.**	**BCI Competition IV**[16]				
Subjects	(FBCSP)	**[12]**	**DC**	**OVR**	**PW**	**[5]**	**[4]**	**1st**	**2nd**	**3rd**	**4th**	**5th**
1	0.68	**0.88**	0.71	0.68	0.78	0.63	0.74	0.68	0.69	0.38	0.46	0.41
2	**0.43**	0.22	0.37	0.42	0.40	0.43	0.43	0.42	0.34	0.18	0.25	0.17
3	0.70	**0.88**	0.66	0.75	0.75	0.74	0.80	0.75	0.71	0.48	0.65	0.39
4	**0.61**	0.39	0.41	0.48	0.52	0.54	0.57	0.48	0.44	0.33	0.31	0.25
5	**0.80**	0.53	0.40	0.40	0.41	0.19	0.31	0.40	0.16	0.07	0.12	0.06
6	**0.42**	0.33	0.26	0.27	0.19	0.26	0.38	0.27	0.21	0.14	0.07	0.16
7	0.55	0.38	0.73	0.77	**0.80**	0.63	0.76	0.77	0.66	0.29	0.00	0.34
8	0.55	**0.85**	0.58	0.75	0.74	0.62	0.75	0.75	0.73	0.49	0.46	0.45
9	0.56	**0.81**	0.50	0.61	0.54	0.69	0.80	0.61	0.69	0.44	0.42	0.37
MKV	**0.59**	**0.58**	**0.52**	**0.57**	**0.57**	**0.53**	**0.61**	**0.57**	**0.52**	**0.31**	**0.30**	**0.29**
stdv.	**0.124**	**0.268**	**0.165**	**0.183**	**0.212**	**0.192**	**0.196**	**0.183**	**0.229**	**0.153**	**0.213**	**0.134**

C. Computational Complexity

In MI classification, classifiers are found to be more computational complex as compared to feature computation and hence dominates the overall computational complexity. In testing, a single input to the classifier can be formulated as $R^{n \times d}$, where R is input sample of dimension $n \times d$. LTCSP method does not consider the inter-class variance and it requires manual selection of weight parameters which is computational expensive as compared to FBCSP which does not require manual selection of weight parameters. SVM and LDA classifiers have $O(max(n,d)min(n,d)^2)$ and $O(nd^2))$ computational complexity respectively. This comparison ensures that the proposed method is less computational costly as compared to the best reported method [4].

IV. CONCLUSIONS

Previous studies have exposed the significance of feature extraction in MI classification. The near-optimal frequency bands and time segment in a trial to extract MI feature also vary from subject to subject. To solve this challenge and to improve classification performance, we have proposed an efficient method to investigate optimal time-frequency range for subject-specific feature extraction. We have combined Band Power (BP) and Time Domain Parameters (TDP) features generated using FBCSP as better features that also minimizes computational complexity in FBCSP-based approaches. The proposed method was evaluated on BCI Competition IV Dataset 2a (four class MI data). Our work has led us to conclude that our method offers better overall performance than other FBCSP approaches. Moreover, our method is robust to artifacts and stable overall 9 subjects and more suitable for real-life applications. As future work, we will combine LTCSP with BP and TDP to extend our experiments toward increase in classification accuracy.

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